llama.cpp Master API Reference

Unified OpenAI-Compatible Documentation for Java Edge Services (2026)

1. Connection Details

Base URL: `http://localhost:8080/v1`

POST /chat/completions

Main endpoint for generating chat responses with support for streaming, tool use, and advanced sampling.

2. Request Schema

Field	Type	Description
`messages`	Array	**Required.** List of message objects `{"role": "user", "content": "..."}`
`stream`	Boolean	Enable Server-Sent Events (SSE). Recommended for Edge UIs.
`temperature`	Float	Randomness (0.0 - 2.0). Default: 0.8.
`min_p`	Float	Minimum probability relative to top token. Highly stable sampler.
`xtc_threshold`	Float	XTC Exclude Top Choices (>0.0). Reduces "AI repetitive" phrasing.
`dry_multiplier`	Float	DRY Don't Repeat Yourself penalty. Prevents loops.
`max_tokens`	Integer	Limit output length. (Alias: `n_predict`).
`samplers`	String	Execution order of samplers (e.g., `"dkypmxt"`).
`cache_prompt`	Boolean	Keeps KV cache in memory for near-instant multi-turn replies.

Field	Type	Description
`grammar`	String	GBNF grammar to force specific formats (JSON/CSV).

3. Payload Examples

Request JSON

```
{ "messages": [ {"role": "system", "content": "You are a helpful agent."}, {"role": "user", "content": "What is 2+2?"} ], "temperature": 0.7, "stream": false, "cache_prompt": true }
```

Response JSON

```
{ "id": "chatcmpl-123", "choices": [{ "index": 0, "message": { "role": "assistant", "content": "2 + 2 is 4." }, "finish_reason": "stop" }], "usage": { "prompt_tokens": 15, "completion_tokens": 8, "total_tokens": 23 } }
```

4. Java Implementation (HttpClient + GSON)

Optimized for edge: Reuses HttpClient, parses JSON into objects via GSON.

```
import com.google.gson.Gson; import java.net.URI; import java.net.http.*; public class LlamaService { private static final HttpClient CLIENT = HttpClient.newHttpClient(); private static final Gson GSON = new Gson(); public String ask(String url, Object payload) throws Exception { var req = HttpRequest.newBuilder() .uri(URI.create(url + "/chat/completions")) .header("Content-Type", "application/json") .POST(HttpRequest.BodyPublishers.ofString(GSON.toJson(payload))) .build(); var res = CLIENT.send(req, HttpResponse.BodyHandlers.ofString()); // Simple GSON navigation var obj = GSON.fromJson(res.body(), com.google.gson.JsonObject.class); return obj.getAsJsonArray("choices").get(0).getAsJsonObject() .getAsJsonObject("message").get(
```

5. Edge Optimization Checklist

- **Memory:** Use `min_p` and `xtc_threshold` to maintain quality at lower bit-depth (4-bit/ Q4_K_M) quantizations.
- **Speed:** Always enable `cache_prompt: true` for agents to avoid re-calculating the entire chat history.

- **Footprint:** Avoid LangChain4j for simple agents; stick to the `HttpClient + GSON` stack to keep .jar size under 1MB.
- **Native:** Use GraalVM to compile your Java service into a native binary for 10ms startup times.